COMP 2521 Winter 2026
Assignment 3: Relational Model and Implementation

This is an individual assignment! No group work is allowed.
You may be asked to explain your work.

Purpose:
To transform the E-R diagrams that were developed during conceptual design into relational database schemas and implement them.

Guidelines:
There are three steps to this assignment. Follow the instructions given below:

Step 1:

- Use the transformation steps used in the tutorial and lectures to transform the given E-R diagram into relations.
- Be sure to underline all the necessary primary keys, include all necessary foreign keys, and indicate referential integrity constraints. **For each relation, indicate the normal forms** it is currently in.
- Diagram the functional dependencies in each relation.

Submit the set of relations using the same format discussed in lectures/tutorials. You may create it using any application (e.g., Excel) along with arrows for functional dependencies / referential integrity constraints or a fully handwritten version. This will be **Version 1**.

Step 2:

- Normalization is a process for reducing anomalies in the resulting relations. Now apply the normalization steps and convert the relations to 3NF.

You will submit the relations in 3NF (you need to show only the ones that require conversion to 3NF). This is **Version 2**.

Step 3:

1. Implement the following relations as MySQL ***tables***.
   i) `model`
   ii) `aircraft`
   iii) `customer`
   iv) `charter`
   v) `employee`
   vi) `crew`
   vii) `credential`

2. Each table will require:

   i) Write CREATE TABLE statements that include all fields specified in the diagram. If an entity has no listed attributes, assign a couple of appropriate ones. For example, the `CREDENTIAL` table can include `cred_id` and `credential_description` to complete it.

   ii) Appropriate domains, primary and foreign keys. When creating tables, a decision needs to be made on the primary key. In some cases, it will be possible to use whatever key was identified in the modeling process; in other cases, you will want to use a 'surrogate' or generated key.

   iii) Populate the tables with sample data, using at least three records for each table listed in 3.1.

   iv) During the initial population, the `yearsInService` field in the `AIRCRAFT` table is left blank. Eventually, in the next step of the assignment, you will be writing a trigger to automatically give this column its value.

You will submit the script file here as `createTest_[MyMRUusername].sql`

Step 4:

1) Write an UPDATE statement to update the `yearsInService` column for all rows using the value in the corresponding `dateOfFirstLaunch` field.

2) Create ***Stored Procedures/Functions*** to do the following:

   a) Add a function getAge (`launchDt`) which returns the age of the aircraft – this function calculates the age (integer) from the `launchDt` passed to it.

   b) Add a new aircraft using the procedure `ADD_AIRCRAFT( modelNbr, aircraftNbr, launchDt)`. First, verify that the `modelNbr` does not already exist in the database; if it does, display an appropriate error message. When adding the aircraft, ensure that records are created in both the `MODEL` and `AIRCRAFT` tables.

   c) Add a procedure `MOD_AC_YEARSERV (modelNbr, aircraftNbr, newLaunchDt)`
   This procedure will change the `dateOfFirstLaunch` for an existing aircraft. If it is NULL, then no change is made.

   d) Add a new record in the table CREW using the stored procedure
   `ADD_CREW (empNbr, charterNbr, cred_desc, role, hrlyRate)`

   e) Ensure that the parameter values passed to the procedure already exist in the corresponding tables before creating a record in the `CREW` table. You may need to retrieve the `cred_id` from the `credential_description` before creating the crew record.

3) Create ***Triggers*** to do the following:

   a) Calculate and store the `yearsInService` from a given `launchDt`, both on:

      i) Inserting a new aircraft record as well as

      ii) Updating the `yearsInService` field of an existing aircraft

      Both of the above triggers use the function getAge (`launchDt`) – written above

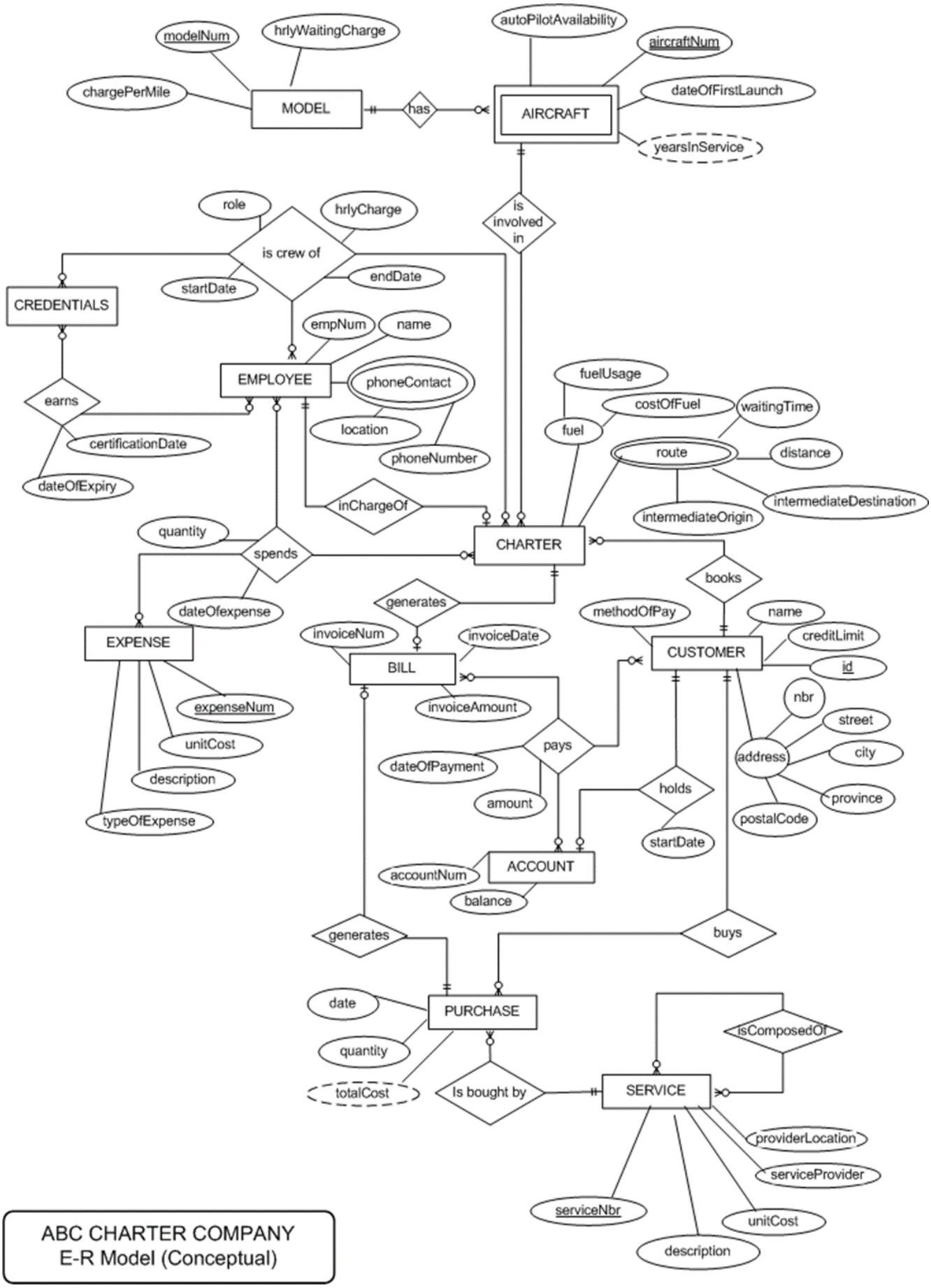
   b) Create an audit table (`crew_charge_audit`) to track changes. Specifically, when the `hrlyCharge` field is modified, record the `old_rate` and `new_rate` for the corresponding `crew_id`, which is the primary key.

4) ***Test***. Use your procedures (and other SQL) to thoroughly test your database. You should have enough testing to thoroughly convince yourself and your instructor that your code works correctly under all situations. Save the test code and submit it as part of the assignment.

Note: For all procedures, when an error condition is detected, a simple error message should be printed, and the operation should not proceed.

Only work that is legible, clean, and professionally presented will be graded. You may require some research to do some parts of the assignment.

You can consult your professor if you need any clarification regarding the assignment.



ABC CHARTER COMPANY
E-R Model (Conceptual)